

Zepto Inventory & Pricing Data Pipeline

Comprehensive SQL Schema, ETL Data Cleaning & Business Intelligence Analytics

- PostgreSQL
- Data Cleaning
- E-Commerce Analytics
- Revenue Optimization

Executive Summary & Domain Context

This technical documentation outlines the database lifecycle for **Zepto's Quick-Commerce Platform**. It covers full DDL table design, exploratory data analysis (EDA), critical sanity cleaning (handling zero prices, currency conversion from paise to rupees), and 8 high-impact SQL business analytical queries targeting revenue estimation, stock movement, inventory categorization, and pricing value metrics.

1. SCHEMA DEFINITION & SETUP (DDL)

Creates the core zepto catalog table structure with appropriate constraints and data types.

```
POSTGRESQL DDL CREATION

DROP TABLE IF EXISTS zepto;

CREATE TABLE zepto (
  sku_id SERIAL PRIMARY KEY,
  category VARCHAR(120),
  name VARCHAR(120),
  mrp NUMERIC(8,2),
  discountPercent NUMERIC(5,2),
  availableQuantity INTEGER,
  discountedSellingPrice NUMERIC(8,2),
  weightInGms INTEGER,
  outOfStock BOOLEAN,
  quantity INTEGER
);
```

2. EXPLORATORY DATA ANALYSIS (EDA)

Initial checks to evaluate dataset size, sample records, missing values, and category distribution.

EXPLORATION OBJECTIVE	SQL QUERY SNIPPET	OPERATIONAL INSIGHT
Total Row Count	SELECT COUNT(*) FROM zepto;	Establishes base catalog volume for verification.
Sample Inspection	SELECT * FROM zepto LIMIT 10;	Validates initial data intake and field formats.
Null Value Check	SELECT * FROM zepto WHERE name IS NULL;	Identifies missing product names requiring remediation.
Distinct Categories	SELECT DISTINCT category FROM zepto ORDER BY category;	Maps unique product verticals in the catalog.
Stock Status Count	SELECT outOfStock, COUNT(sku_id) FROM zepto GROUP BY outOfStock;	Measures catalog availability vs stockout risks.

EXPLORATION OBJECTIVE	SQL QUERY SNIPPET	OPERATIONAL INSIGHT
Duplicate SKU Names	<code>SELECT name, COUNT(sku_id) FROM zepto GROUP BY name HAVING COUNT(sku_id) > 1;</code>	Detects redundant listing entries for deduplication.

3. DATA CLEANING & NORMALIZATION

Purging corrupted records and re-scaling currency figures from implicit paise units into standard INR rupees (₹).

DATA CLEANING PIPELINE

```
-- 1. Identify corrupted listings with zero price
SELECT * FROM zepto
WHERE mrp = 0 OR discountedSellingPrice = 0;

-- 2. Remove corrupted zero-price items from table
DELETE FROM zepto
WHERE mrp = 0;

-- 3. Currency Normalization: Convert paise values to rupees (÷ 100.0)
UPDATE zepto
SET mrp = mrp / 100.0,
    discountedSellingPrice = discountedSellingPrice / 100.0;

-- Verification query
SELECT mrp, discountedSellingPrice FROM zepto;
```

4. BUSINESS ANALYTICS & INTELLIGENCE QUERIES

Targeted queries answering key commercial questions for inventory, pricing strategies, and revenue forecasts.

Q1. Top 10 Best-Value Products by Discount Percentage

Identifies products offering maximum markdown percentage to feature in promotional campaigns.

```
SELECT DISTINCT name, mrp, discountPercent
FROM zepto
ORDER BY discountPercent DESC
LIMIT 10;
```

Q2. High MRP Products Currently Out of Stock (MRP > ₹300)

Highlights lost revenue opportunities on premium items currently unavailable for order fulfillments.

```
SELECT DISTINCT name, mrp
FROM zepto
WHERE outOfStock = TRUE AND mrp > 300
ORDER BY mrp DESC;
```

Q3. Estimated Inventory Revenue per Category

Calculates total potential revenue tied up in current inventory: $Revenue = \sum (SellingPrice \times AvailableQty)$

```
SELECT category,
       SUM(discountedSellingPrice * availableQuantity) AS total_revenue
FROM zepto
GROUP BY category
ORDER BY total_revenue DESC;
```

Q4. Premium Products with Low Discount (MRP > ₹500, Discount < 10%)

Identifies high-value items sold with minimal margins/discounts, suitable for margin optimization or retention.

```
SELECT DISTINCT name, mrp, discountPercent
FROM zepto
WHERE mrp > 500 AND discountPercent < 10
ORDER BY mrp DESC, discountPercent DESC;
```

Q5. Top 5 Categories Offering Highest Average Discount Percentage

Pinpoints the product categories driving price competition and aggressive customer acquisition discounts.

```
SELECT category, ROUND(AVG(discountPercent), 2) AS avg_discount
FROM zepto
GROUP BY category
ORDER BY AVG(discountPercent) DESC
LIMIT 5;
```

Q6. Price Per Gram Analysis for Items ≥ 100g

Computes unit economy cost metric: $PricePerGram = SellingPrice \div WeightInGms$ sorted for consumer value.

```
SELECT DISTINCT name, weightInGms, discountedSellingPrice,
       ROUND((discountedSellingPrice / weightInGms), 2) AS price_per_grm
FROM zepto
WHERE weightInGms >= 100
ORDER BY price_per_grm ASC;
```

Q7. Weight Classification Tiering (Low, Medium, Bulk)

Segments inventory items into operational delivery buckets based on unit weight boundaries.

```
SELECT DISTINCT name, weightInGms,
       CASE
         WHEN weightInGms < 1000 THEN 'Low'
         WHEN weightInGms < 5000 THEN 'Medium'
         ELSE 'Bulk'
       END AS weight_category
FROM zepto;
```

Q8. Total Inventory Weight Load per Category

Calculates aggregated warehouse weight burden: $TotalWeight = \sum (WeightInGms \times AvailableQty)$ for logistics capacity planning.

```
SELECT category,  
       SUM(weightInGms * availableQuantity) AS total_weight  
FROM zepto  
GROUP BY category  
ORDER BY total_weight DESC;
```